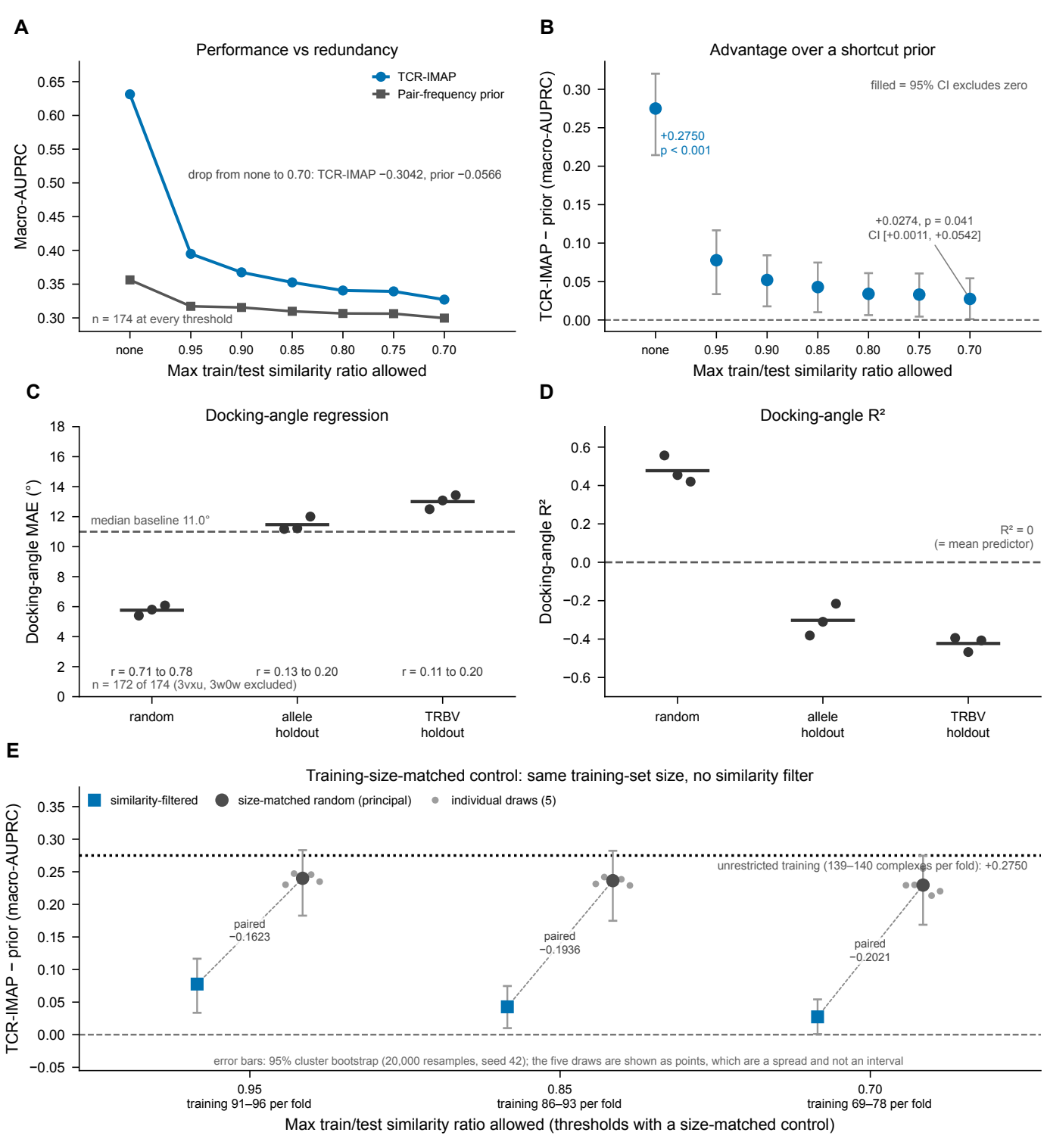
A
Performance vs redundancy
TCR-IMAP
Pair-frequency prior
drop from none to 0.70: TCR-IMAP −0.3042, prior −0.0566
n = 174 at every threshold
Macro-AUPRC
Max train/test similarity ratio allowed
none
0.95
0.90
0.85
0.80
0.75
0.70
B
Advantage over a shortcut prior
filled = 95% CI excludes zero
+0.2750
p < 0.001
+0.0274, p = 0.041
CI [+0.0011, +0.0542]
TCR-IMAP − prior (macro-AUPRC)
Max train/test similarity ratio allowed
C
Docking-angle regression
median baseline 11.0°
r = 0.71 to 0.78
r = 0.13 to 0.20
r = 0.11 to 0.20
n = 172 of 174 (3vxu, 3w0w excluded)
Docking-angle MAE (°)
random
allele holdout
TRBV holdout
D
Docking-angle R²
R² = 0
(= mean predictor)
Docking-angle R²
random
allele holdout
TRBV holdout
E
Training-size-matched control: same training-set size, no similarity filter
similarity-filtered
size-matched random (principal)
individual draws (5)
unrestricted training (139–140 complexes per fold): +0.2750
paired −0.1623
paired −0.1936
paired −0.2021
error bars: 95% cluster bootstrap (20,000 resamples, seed 42); the five draws are shown as points, which are a spread and not an interval
TCR-IMAP − prior (macro-AUPRC)
0.95
training 91–96 per fold
0.85
training 86–93 per fold
0.70
training 69–78 per fold
Max train/test similarity ratio allowed (thresholds with a size-matched control)